SINGAPORE DIMENSIONS • GRADE 1A • CHAPTER 2

Number Bonds to 10 & Complements

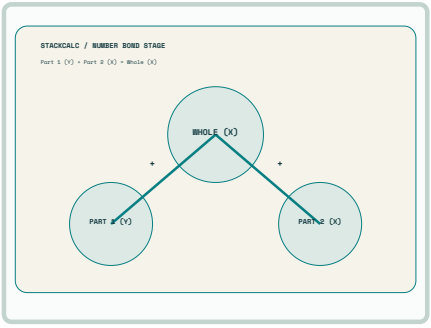
Tactile Concrete-Pictorial-Abstract (CPA) Station Guide • CCSS.MATH.1.OA.C.6

Student Name: _____ Date: _____ **Model: SC-6** **Check Score: ___ / 4**

STUDENT SELF-CHECK RULE (NO CHEATING):
Build with blocks and solve on paper first! Use your SC-6 calculator ONLY to verify answers on the 4-line LCD screen.

STATION LAB TUTORIAL: BENCH SCHEMATIC & CPA PROGRESSION

1. 2D Bench Blueprint



Kit: Number Bond Stage

2. Desk Setup (Concrete -> Pictorial)

Physical Bench Setup:
Place 7 counters on the 10-frame. Count empty cells to find complement.

Sketch Model on Paper:
Draw 10-frame with 7 shaded dots and 3 open circles.

3. Step-by-Step LCD Stack Trace (SC-6)

MODE: 123 Base-10 Arithmetic

1.	1	Step 1	Execute [1]
2.	0	Step 2	Execute [0]
3.	ENTER	Step 3	Execute [ENTER]
4.	8	Step 4	Execute [8]

EXPECTED LCD X: 2

1 **What is the complement of 6 to make 10?**

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer: _____

☐ Verified on SC-6

2 **What is the complement of 4 to make 10?**

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer: _____

☐ Verified on SC-6

3 **What is the complement of 9 to make 10?**

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer: _____

☐ Verified on SC-6

4 **What is the complement of 1 to make 10?**

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer: _____

☐ Verified on SC-6

Station Diagnostic & Solution Key: Chapter 2

Number Bonds to 10 & Complements • Manipulative Setup, Diagnostic Rubric & Verification Flow

MANIPULATIVE HARDWARE STATION INVENTORY & 3D PRINT PREPARATION

Station Kit: Ten-Frame Studio & Math Dominoes

CAD Blueprint: number_bond_stage

Target Device: SC-6 Elementary

CAD / STL Source File:

Education/scad/math_dominoes_parts.scad

Standard: Gridfinity 42mm Pitch / 0.6mm Inset Markings

Recommended Filaments:

Oceanic Teal & Warm White (PLA)

#	Math Problem Prompt	Expected Math Answer	Self-Check Keystrokes & RPN Stack
1	**What is the complement of 6 to make 10?** Standard: CCSS.MATH.1.OA.C.6	4	[1][0] [ENTER] [6] [-]
2	**What is the complement of 4 to make 10?** Standard: CCSS.MATH.1.OA.C.6	6	[1][0] [ENTER] [4] [-]
3	**What is the complement of 9 to make 10?** Standard: CCSS.MATH.1.OA.C.6	1	[1][0] [ENTER] [9] [-]
4	**What is the complement of 1 to make 10?** Standard: CCSS.MATH.1.OA.C.6	9	[1][0] [ENTER] [1] [-]

TEACHER DIAGNOSTIC GUIDE: COMMON MISCONCEPTIONS & REMEDIATION

Key Misconception to Target During Station Observation:

Ensure students internalize 10-complements (1/9, 2/8, 3/7, 4/6, 5/5) for mental math.

How the Manipulatives & Calculator Foster Self-Correction:

Students solve with pencil and physical blocks first. When they self-check on the SC-6, any discrepancy between their written answer and the calculator's 4-line LCD immediately prompts them to re-inspect their physical bench model or regrouping, turning errors into productive struggle without teacher intervention.

Station Facilitation & Pacing Tip:

Keep 3D printed trays docked on standard Gridfinity baseplates to prevent loose parts on student desks. Encourage students to articulate their CPA steps: name the block representation (Concrete), point to their drawn tape/bond model (Pictorial), and verify register values (Abstract).

CLASSROOM SCORING & STATION MASTERY RUBRIC

Exemplary (18–20 pts):	Complete, accurate manipulative sketches; exact math answers; independently verified self-check on LCD stack.
Proficient (14–17 pts):	Correct math answers; minor sketch omissions; verified on calculator with minimal teacher prompting.
Developing (10–13 pts):	Understands problem concept but made computational slips; successfully self-corrected using calculator LCD.
Intervention (<10 pts):	Relied on calculator before solving on paper, or struggled with underlying physical manipulative models.